

Shakespeare AI: Token Usage & API Credit Breakdown

Technical Analysis of LLM Call Metrics and Optimization Policies

Executive Summary: This document provides an audit of the token usage profiles associated with the Shakespeare Speaks multi-agent application. It details how user queries, background loops, and long-form analysis features translate into Vertex AI / Gemini API token credit consumption, helping developers analyze and optimize billing overheads during testing and production stages.

1. The Essay / News Analysis Page

When a user triggers a Shakespearean commentary or analysis on a current events or climate story card, the application performs a long-form generation task. This page is the highest token-consuming user action per request due to the length of the generated output.

- **Input Safety Check:** A pre-flight model call is made to inspect the headline and excerpt text to detect and block prompt injections. *(Token count: ~200 input, ~10 output).*
- **Analysis Generation Call:** The model is invoked with the headline, excerpt, and the matched quote, instructing it to write a detailed, multi-paragraph Markdown analysis in a serene, philosophical persona. *(Token count: ~1,000 input, ~800 to 1,500 output).*

2. The Conversational Chat Page

The chat interface coordinates multi-agent intent routing through the ADK runner. Every message submitted by a user goes through a three-step processing pipeline:

1. **Input Safety Check:** Runs the pre-flight guardrail check on the input message. *(Token count: ~200 input, ~10 output).*
2. **Intent Routing:** The ADK router classifies the input against a strict Pydantic schema to direct it to the correct subagent. *(Token count: ~800 input, ~30 output).*
3. **Subagent Execution:** The targeted agent processes the message:
 - *Advice/General Chat:* Simple prompt-to-response generation. *(~1,000 input, ~300 output).*
 - *History Agent:* Reads local Markdown files as context. *(~2,000 to 4,000 input, ~300 output).*
 - *Scholarship Agent:* Calls Google Search Grounding to fetch live web contexts. *(~3,000 to 5,000 input, ~300 output).*

3. The News & Climate Background Caching Loop

To maintain homepage cards, a parallel background refresher loop checks the local JSON cache files. Every 12 hours (configured for Vertex pre-live testing, returning to 15 minutes in production), the background worker refreshes the cache if stale:

- **Step 1 (Grounded Search):** A call is made with Google Search Grounding to collect the latest top news. *(Token count: ~1,500 input & output).*
- **Step 2 (JSON Parser):** Feeds the search text into a schema-constrained structured generator to parse it into 10 clean story models with quotes. *(Token count: ~3,000 input, ~1,500 output).*

4. Local Non-Token Operations (Auth and Telemetry)

Endpoints related to user registration, login, and telemetry tracking (such as mobile scroll events or page loads) are executed entirely locally in SQLite using cryptographic PBKDF2 hashing. These operations consume **zero LLM tokens**.

Summary Token Metrics Table

Feature / Endpoint	Input Tokens	Output Tokens	API Grounding Extra
Safety Pre-flight	~200	~10	No
ADK Intent Routing	~800	~30	No
Essay/News Analysis	~1,000	~800 - 1,500	No
Chat (General / Advice)	~1,000	~300	No
Chat (History Agent)	~2,000 - 4,000	~300	No (Local Files)
Chat (Scholarship Agent)	~3,000 - 5,000	~300	Yes (Google Search)
Cache Refresher (Per Loop)	~4,500	~3,000	Yes (Google Search)